

OENG1250 Intro to Engineering Design

Assessment Task-2 : Group Design Project

Lunar Rover Project

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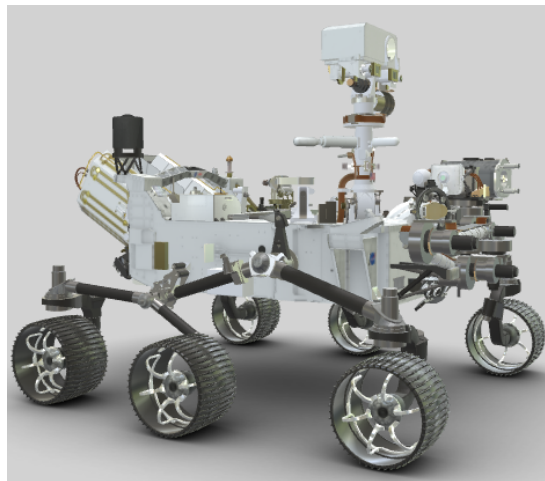


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1 Assessment Summary

Assessment 2 for this course consists of a semester long project with a combination of individual and group tasks, contributing a total of 50% to your final course mark. Students will work in a team of 6 for both Part A and B of the project. All submissions are due on Sunday at 11:59 pm of the due week given in Table 1.

Table 1: Assessment 2: Group Design Project assessment details

Component	Type	Mark	Due*	Submission	Notes
Part A	Individual	27%	Week 8	Individual	Assessed individually. Submission requires some items from group activity for Rover sub-assembly.
Part B	Group (Team of 6)	20%	Week 12	Group	Assessed as group. Individual mark scaled for lack of group participation.
Project Quiz 1	Online Quiz	1%	Week 2	Individual	Multiple attempts allowed.
Project Quiz 2	Online Quiz	1%	Week 5	Individual	Multiple attempts allowed.
Project Quiz 3	Online Quiz	1%	Week 10	Individual	Multiple attempts allowed.
Total	–	50%	–	–	Total course weighting

**Refer to Canvas or the RMIT Higher Education Important Dates page using this link ([Important Dates](#)) to find due dates and time.*

Part A is an individual submission requiring some items completed as group activity towards your group's Rover design. Thus students are required to work with their group from Week 4 to 8 to receive marks for **Part A**. Refer to the detailed explanation of **Part A** submission requirements given in Section 5 on pages 8-9

Part B is a **group submission** by a single member on behalf of the entire team. Although marked as a group, individual scores may be adjusted based on each team member's level of participation in the group work measured from Week 4 to Week 12.

In addition, there are three **Project Professional Practice Quizzes**, each an online quiz worth 1%, due in **Weeks 2, 5, and 10**, respectively. These quizzes are individually assessed for each team member on Canvas, and **multiple attempts are allowed**.

1.1 Step-by-Step: Process for Forming and Joining a Group

- Form a group with up to 6 students**
 - Talk to classmates in your **tutorial class** and agree to work together.
 - Groups can also be formed across different tutorial sessions or lectorials if all members agree to collaborate regularly.
 - Once your team has found 5 or 6 members, you can all join a group on Canvas.
- Find a pre-created group on Canvas**
 - Go to the **People tab** on Canvas.
 - Scroll down to the list titled “**Group 1 Rover Project**” to “**Group 290 Rover Project.**”
- Join a group on Canvas**
 - Each member must **manually join the same group** pre-created by the teaching team.
 -  Only join a group that has **fewer than 6 members**.
 -  **Do not click the “+Group” button**, this creates a new group and will not be recognised.
- Deadline to join a group:**
 -  **Friday of Week 3 at 5:00 PM**
 - After this, the system will be locked and no further self-joining will be allowed.
 - Unassigned students will be placed into groups by the teaching staff at the start of Week 4.
- Refer to Canvas [FAQ page](#) questions “What is the process and deadline for joining a group...?”

2 Project Background

The \$150 million Australian Moon to Mars initiative supports Australian businesses and researchers to join the National Aeronautics and Space Administration's (NASA) inspirational endeavour to go forward to the Moon and then go on to Mars. The Australian Space Agency has funded a consortium of engineering companies and research organisations for this initiative. You are working in a student design team at the RMIT Space Industry Hub which is part of this consortium. Your team has been tasked with designing a lunar rover (the Rover) capable of collecting lunar regolith and delivering it to a NASA science in-situ resource utilisation (ISRU) facility on Moon. The rover will fly on a NASA mission to lunar surface in 2026. The mission involves several phases including pre-launch, launch, cruise and orbit to Moon, landing, post-landing egress and rover on lunar surface phases. Your team must create a functional design of the rover that meets the mission objectives and requirements for the preliminary and detailed design review phase of the project. The successful design will progress to the next phase involving, design analysis and safety review, critical design review, acceptance and qualification review, flight and operational readiness review, which is beyond the scope of this assessment task.

3 Rover Mission on Lunar Surface

The rover has one primary task, to delivery lunar regolith to the ISRU facility (F) shown in Figure-1. To achieve this, the rover will need to survey the environment, find an acquisition site, acquire the regolith, transport it to the regolith delivery zone and deliver it to the ISRU facility. The rover will repeat this delivery task until the requirements are met. Other tasks include:

- repeated loading and unloading of lunar regolith
- constructing regolith stores or stockpiles
- conducting remote asset inspection
- performing surface preparation - terrain assessment, shaping and/or rock clearing.
- end of life task into the permanently shadowed regions.

The rover will finally lay to rest when the mission is complete and the sun sets on its location.

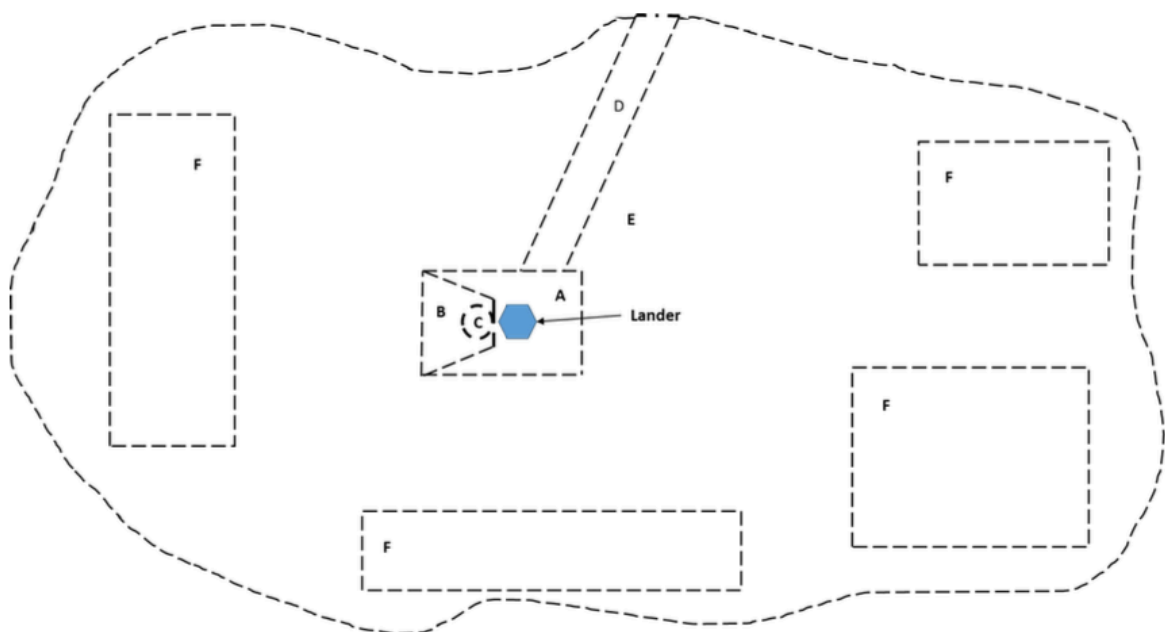


Figure-1 : An example diagrammatic representation of Mission Operational Zones on the lunar surface. The regolith acquisition site is expected to be within 300 m of the regolith deposit sites F.

4 Rover Design Requirements

4.1.1 Functional Requirement

- The Rover shall acquire lunar regolith from the target acquisition site and deliver to ISRU.

4.1.2 Performance Requirement

4.1.2.1 Mass

- The Rover mass shall not exceed 20 kg

4.1.2.2 Volume

- The rover dimensions shall not exceed 300mm x 500mm x 500mm (Height x Width x Length), (height dimension is perpendicular to lunar surface)

4.1.2.3 Operation

- The rover shall operate for a minimum of one standard lunar daylight period (354 hours)
- The rover shall operate at a minimum speed of 0.28 km/hr and maximum speed of 0.37 km/hr

4.1.2.4 Regolith Amount

- The Rover shall deliver 1 kg of regolith in 5 days timeframe.
- The Rover shall deliver at a minimum of 100 g of regolith in 1 day timeframe.
- The Rover shall deliver at a maximum of 1 kg of regolith in 1 hour timeframe.
- The delivered lunar regolith shall contain particles no larger than 100 cm³.
- The delivered lunar regolith shall not contain particles with any single dimension less than 2 cm or greater than 5 cm.
- The delivered lunar regolith shall have a density of approximately 1.5 g/cm³

4.1.2.5 Terrain and Ground Clearance

- The Rover shall have onboard camera mounted on a mast for surveying the terrain.
- The Rover shall have a minimum ground clearance (i.e. distance between lunar surface and underside of rover measured between the end points for the drive system motor or strut) of 200 mm.
- The Rover shall have a minimum of 6 wheels for navigating the challenging lunar terrain.
- The Rover shall have wheels with a minimum diameter of 80 mm for navigating the challenging lunar terrain.
- The Rover shall have wheels with a maximum diameter of 130 mm for navigating the challenging lunar terrain.
- The rover suspension shall feature a single rocker-bogie design for the front wheel.
- The rover suspension shall feature a dual rocker-bogie design for the rear wheels.

4.1.2.6 Power

- The Rover shall operate on nominal power of 20 W
- The Rover shall have onboard 12 V solar panel rated at 20 W with a dimension of 200mm x 300mm (Height x Width x Length), and weighing 2.15 kg for recharging operations.
- The Rover shall have an onboard 12 V power pack designed from individual Li-ion 18650 cells, which has an energy density of 250 Wh/kg (gravimetric) or 559 Wh/l (volumetric).

4.1.2.7 Drive System

- The Rover shall have independent propulsion for each wheel.
- The Rover shall operate on 12V DC powered electric motor certified for Lunar operation.
- The Rover must use the 12V DC powered electric motor model shown in Section 4.2.

4.1.2.8 Robotic Arm Mount

- The Rover shall provide attachment points for fixing the mounting plate of robotic arm to be used for regolith collection and manipulation, which will be designed by a separate team. The robotic arm component will weigh 2.0 kg and will be assembled with the Rover by NASA.
- The mounting plate for the robotic arm shown in Figure 2 has an outer diameter (OD) of 150 mm, featuring 4 x M10 bolts at a pitch circle diameter (PCD) of 120 mm.
- Teams are only required to provide attachment points, and not the design of a robotic arm.

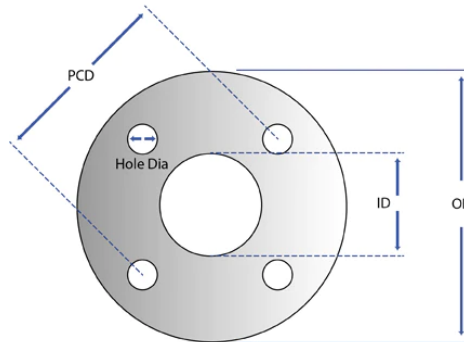


Figure 2: Design details of the robotic arm mounting plate with an OD = 150 mm to be fixed to the rover. Use 10 mm hole diameter for M10 bolt with a clearance fit, and PCD of 120. No need to include the centre hole with ID.

4.2 Rover Components

4.2.1.1 The Rover-Assembly design must include the following rover components. Each member of the team will be responsible for designing one component:

1 Wheels

Outer diameter ranging between 80 mm to 130 mm. Final diameter must be calculated based on the team's mission planning for collection of regolith. Can be designed as a single part or assembly of multiple parts. Recommended material for estimation of weight include, aluminium, titanium, carbon-fibre polymer composite, or steel.

2 Suspension

Must be designed as an assembly from multiple suspension links with a single rocker-bogie design for the front wheel and a dual rocker-bogie design for the rear wheels. It is not necessary to provide detailed design of the connections such as bearing, bushing, or anti-roll bars. Recommended material for estimation of weight include, aluminium, titanium, carbon-fibre polymer composite, or steel.

3 Body

Must provide attachment points for Suspension, Robotic Arm (see Figure 2), Camera Mast, and housing for Powerpack and Solar Panel. Must provide a container that is big enough for housing regolith collected in a single trip. Can be designed as a single part or assembly of multiple parts. Recommended material for estimation of weight include, aluminium, titanium, carbon-fibre polymer composite, or steel.

4 Drive System

The Drive System must be designed using an assembly of electric motor and strut for connecting the motor to the suspension link. It must feature the electric motor certified for Lunar operation shown in Figure 3. The Rover electric motor weighs 300 g and it can operate with maximum rated torque at a power of 3 W with the shaft speed of 10 RPM. Recommended material for estimating the weight of strut include, aluminium, titanium, carbon-fibre polymer composite, or steel.

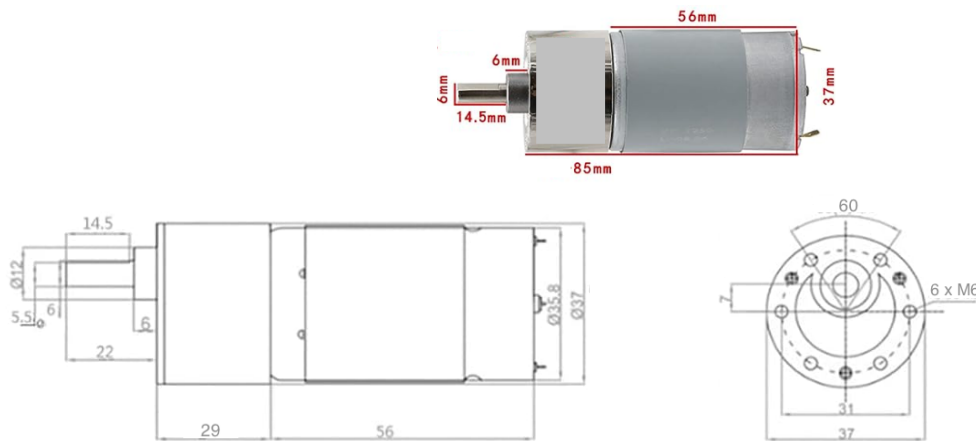


Figure 2: Design details of the electric motor for the rover drive system. 6 x M6 mounting plate fasteners can be provided at the front or rear face of the motor.

5 Powerpack and Solar Panel

Design a 12V DC power supply using Li-ion 18650 cells, which has an energy density of 250 Wh/kg (gravimetric) or 559 Wh/l (volumetric). Must consist of an assembly of individual cells contained within the housing, e.g. a pack of 40 individual cells. The 18650 lithium ion cell shall be cylindrical shaped with the following dimensions: Diameter = 18.2 mm and Height = 65.10 mm, as shown in Figure 2. Teams can exclude the details of the contact point and cutout seen on the cell in Figure 2. Recommended material for estimating powerpack weight include, lithium-ion with mass density of 2700 kg/m³. The Rover shall have onboard 12 V solar panel rated at 20 W with a dimension of 20mm x 200mm x 300mm (Height x Width x Length), and weighing 2.15 kg for recharging operations (i.e. specific power of 9.2 W/kg). Solar panel can be designed as a single part with minimal design details. [Assign the design task for Powerpack and Solar Panel to the fifth member of your team who is less active or inactive.](#)

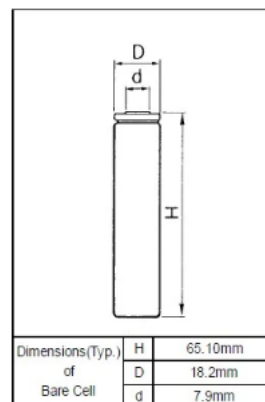


Figure 2: Dimensions of a single 18650 lithium ion cell to be used in the power pack.

6 Mast with Camera Housing

Design a camera housing to suit a small GoPro-sized camera measuring 55 mm x 55 mm x 55 mm. A camera system will be acquired off-the-shelf weighing 130 gm. The mast shall feature a telescopic design that can extend a minimum length of 125 mm. The maximum allowable height for the Rover is measured with the camera mast retracted. [If your team has 5 members, this rover component is not required in your final Rover assembly design. If there are 6 members, this component design task must be assigned to the 6th member only or a member who is not responding to group meeting requests or is inactive in the project\).](#)

5 Part A – Individual Submission

Due Date: Week-8, 11:59 pm Sunday, Weight: 27% of course mark. Feedback returned in Week-10

Objectives: Students must work in a team of six, with each member responsible for designing one of the six rover components listed in Section 4.2. using Solidworks 2023. Each student is required to maintain an individual design journal, documenting the entire design process, from initial ideation and concept generation through sketching, to preliminary and detailed design development using CAD modeling. Team collaboration is essential. Students must actively participate in group meetings to agree on critical design features, particularly those that connect with adjacent/connecting components being developed by other team members. To support this collaboration, each student must share their preliminary design model of the assigned rover component with the team by Week 6, enabling the successful development of the Rover Sub-assembly.

Table-2 : Part-A marks distribution, submission requirements and expected milestone completion date.

No.	Submission Item	Description	Marks	Milestone
1.1	Group Formation	Team of 6 formed via Canvas; each student assigned one rover component	-	Week 3
1.2	1st Group Meeting	Schedule group meetings and assign rover component to each member. Setup a shared file for recording weekly meeting minutes and group design journal.	-	Week 4-5
2.	Model Development	Submit a single .zip folder titled StudentNumber_Name.zip with the required files		
2.1	Preliminary Design Model	Create a simple CAD model of your component with basic geometry and connection features for early assembly testing. Design can be same or different from examples demonstrated in the class. Model can be a single part or assembly.	10%	Week 5-6
2.2	Rover Sub-assembly Model	Assemble your preliminary component with adjacent/connecting components of the rover to demonstrate correct fit and connectivity; share Preliminary Design model files with group by Week 6.	10%	Week 6-7
2.3	Detailed Design Model	Create a high-fidelity 3D model with refined dimensions, geometry, detailed assembly features, and updated connection/interface design based on group feedback. Model must be an assembly and an original design demonstrating improvement on the design examples provided in the class. Note, Wheel or Body can be a single part.	15%	Week 8
3.	Individual Design Journal	Include Individual Design Journal as a single .PDF file in StudentNumber_Name.zip submission		
3.1	Contribution table	Include Part A Group Work Contribution Table signed by all team members. Example provided in <i>Appendix 3 Table 1</i>		
3.2	Problem Statement	For your assigned component, clearly describe the design objectives, functional requirements, and dimensions and constraints provided to you by the team.	1%	Week 4
3.3	Brainstorming	Document ideation process using sketches, concept alternatives, selection rationale, calculation or analysis and progression toward CAD implementation.	4%	Week 4-5
3.4	Meeting Minutes	Maintain records from at least 5 group meetings including attendees, agenda, design decisions, responsibilities, and shared deadlines. See Appendix 3 for template.	10%	Week 3-8
3.5	Share Preliminary Design Model	Submit and share your preliminary design CAD model with the team by end of Week 6 to allow rover sub-assembly testing by the group.	10%	Week 6
3.6	Technical Drawing - Preliminary Design	Using Solidworks, generate technical drawing of preliminary model including all orthographic views, an isometric view, full dimensions, scale and title block.	10%	Week 6
3.7	Technical Drawing - Detailed Design	Generate complete technical drawing of the detailed model. It contains multiple pages and communicates both part and assembly features. It must include: Page 1: Exploded assembly view of parts with Balloons and a Bill of Materials (BOM), clearly arranged and numbered. Pages 2–4: Orthographic, sectional, and isometric views for each part used in the rover component (e.g. front, middle, and rear suspension links), showing all dimensions, title blocks, and annotations.	20%	Week 7
3.8	Rover Sub-assembly Drawing	Create exploded view drawing of your rover sub-assembly showing assembly compliance with adjacent/connecting rover components. It must include: Page 1: Exploded assembly view of rover components with Balloons and a BOM, arranged clearly. Page 2: Orthographic and isometric views of the rover sub-assembly with major dimensions only (e.g. height, width, length), scale, and title block.	10%	Week 8

5.1 Part-A Individual Submission Requirements

Students must submit a **single .zip folder** titled “StudentNumber_StudentName.zip”. For example “3212302_John Smith.zip”. Submitted **.zip** folder must contain following sub-folders and associated item listed in Table 3.

Table 3: Part A submission (StudentNumber_Name.zip folder) must contain sub-folders and files types

Sub-folder Name	File Format	Submission Item
Preliminary Design Model	.SLDPRT or .SLDASM include all relevant .SLDPRT files) and .STEP	CAD file of your preliminary design model. If using an assembly, ensure all individual part files are included.
Detailed Design Model	.SLDASM (include all relevant .SLDPRT files) and .STEP	CAD file of your detailed design model. If using an assembly, ensure all individual part files are included.
Rover Sub-assembly Model	.SLDASM (include all relevant .SLDPRT files of adjacent parts) and .STEP	Sub-assembly model showing connection to adjacent components using the preliminary design. Submit with required part files.
Design Journal (PDF format)	.PDF	Include your Individual Part Design Journal. Must contain <i>Table 1 from Appendix 3</i> , showing group work contributions signed by all team members

5.2 Refer to [Frequently Asked Questions for Assessment 2 Part A \(link\)](#) on Canvas for an explanation of submission requirement with examples and demonstration videos

- *Q: What files do I need to submit, and what is their format for Part-A Individual Design due in Week-8?*
- *Q: How to Save Solidworks PART or ASSEMBLY files correctly for Assignment Submission and Sharing?*

6 Part B – Group Submission

Due Date: Week-12, 11:59 pm Sunday, Weight: 20% of course mark. Feedback returned in Week-13.

Objectives: Group must develop assembly model of the Rover from the rover components designed by each team member during Part-A. Students must share the detailed design model for their rover component from Part A with the team by no later than Week 8 or 9 for completion of the rove assembly. Group must create a 10 minute video presentation explaining the process used to achieve the Rover design outcome from ideation through to implementation. Demonstrate that the final design meets objectives and requirements. Explain the key features of your design using assembly model, technical drawings, virtual reality model, etc.

Table-4 : Part B marks distribution, submission requirements and expected milestone completion date:

No.	Submission Item	Description	Marks	Milestone
1.	CAD Assembly Model of Rover	Develop a full CAD assembly model of Rover using the Detailed Design model of rover components created by each team member in Part A.	10%	Week 9–10
		Include exploded view of the assembly model with accurate alignment and direction of assembly that clearly communicates the relationship between individual components.	10%	Week 11
		Include an animation of the Rover assembly and disassembly sequence using the correct sequence of steps, with accurate positioning and alignment of all individual components for supporting manufacturing process.	10%	Week 11
2.	Video Presentation	Create a 10-minute video presentation explaining the entire design process, from initial concept to final implementation. Must include the items listed below.	-	Week 12
2.1	Design Process and Outcome	Explain how the design evolved from ideation through to implementation. Show how the final design meets all specified objectives and requirements.	15%	Week 9-11
2.2	Design Features	Present key design features using CAD models and drawings. Highlight projections, exploded views, and part assembly sequences to engage expert reviewers.	20%	Week 12
2.3	Technical Drawings	Present full technical drawings of the rover assembly, including, exploded views, BOM, 3-view projections with key dimensions (e.g. overall width, length, height with camera mast retracted), scale, and title block.	5%	Week 10
2.4	Virtual Reality Demo	Create a VR demo showing how the rover would function in its intended operational environment.	10%	Week 11–12
2.4	Presentation Skills	Each student participated in the video presentation.	10%	Week 12
3.	Group Design Journal	Document collaborative team work and group processes throughout the project duration.	—	—
3.1	Brainstorming	Document ideation process using sketches, concept alternatives, selection rationale, calculation or analysis and progression toward CAD implementation.	5%	Week 4-9
3.2	Meeting Minutes	Record participants, agendas, design discussions, file sharing, deadlines, and action items from team meetings held throughout the project.	5%	Week 4-9

6.1 Part-B Group Submission Requirements

One student from the group must submit a single .zip folder titled “GroupNo.zip”. For example “Group123.zip”. If you name, student ID and signature is included in Table 2a and 2b, group member contribution to Part B given in Appendix – IV, your Part B will be marked as submitted.

Table 5: Part B submission (GroupNo.zip folder) must contain sub-folders and files types

Sub-folder Name	File Format	Submission Item
Assembly Model	.SLDASM (include all relevant .SLDPRT files) and .STEP	Final assembly model of the complete rover. Must include all associated part files.
Video Presentation	.MP4	Recorded group presentation video outlining the design process and outcomes, features, compliance, VR demo, assembly animation, etc
Group Design Journal (PDF format)	.PDF	Group Design Journal. Must include Tables 2a and 2b, and details of each member’s contribution to the project as per Appendix IV.

7 Appendix 1 – Definition of CAD Model and Drawing

7.1 Part Model:

A *Part Model* is a single CAD part file created in SolidWorks with the extension `.sldprt`. It represents an individual component with no assembly features. Part Models are the basic building blocks used to construct assemblies or sub-assemblies.

7.2 Assembly Model:

An *Assembly Model* is a SolidWorks CAD file with the extension `.sldasm` that combines two or more Part Models (`.sldprt`). Assemblies may also include other assemblies, and can be referred to as *sub-assemblies* when used as part of a larger system such as a Rover, Robot, Car, or Aircraft.

7.3 Rover Component:

A *Rover Component* refers to a complete CAD model representing one of the key functional systems of the rover (as listed in Section 4.2). It may be:

- a single part (e.g. Wheel, Body), or
- an assembly (e.g. Suspension, Drive System, Powerpack, Camera Mast).

7.4 Rover Sub-assembly:

A *Rover Sub-assembly* is a CAD Assembly Model created by joining 2 or more Rover Components. It is used to verify compatibility and correct alignment of connection features before final assembly. In Part A, each student will create a sub-assembly based on their assigned component:

1. **Wheel Designer:** Wheel assembled with Drive System
2. **Drive System Designer:** Drive System assembled with Wheel and Suspension
3. **Suspension Designer:** Suspension assembled with Drive System and Body
4. **Powerpack and Solar Panel Designer:** Powerpack and solar panel assembled with Body
5. **Camera Mast Designer:** Mast with camera housing assembled with Body
6. **Body Designer:** Body assembled with Suspension (one side only), Powerpack, Solar Panel, and Camera Mast — excluding any parts not developed by the team.

7.5 Preliminary Design Model:

A *Preliminary Design Model* is the initial CAD model (Part or Assembly) of a Rover Component, created using minimum required design effort. It must:

- include all key dimensions based on the mission requirements,
- clearly show connecting features needed to interface with adjacent connecting rover components (i.e. Component B to A and C, see section 5.4),
- be developed collaboratively with the team and revised as needed.

This model can be same or different from tutorial or lectional examples. In most cases, the features of this model may be unsatisfactory to the team, thereby requiring further refinement which will be included in the Detailed Design Model.

7.6 Detailed Design Model:

A *Detailed Design Model* is the final version of a Rover Component's CAD model (Part or Assembly), incorporating:

- all mission-based dimensions and any design changes agreed upon by the team,
- complete connecting and assembly features to integrate with adjacent components,

- improvements or refinements beyond tutorial examples.

This model must meet the final design requirements and demonstrate higher quality to earn full marks.

7.7 Rover Assembly

The *Rover Assembly* is the final full CAD assembly model combining all components listed in Section 4.2. It must be submitted in **Part B (Week 12)**.

7.8 Technical Drawing of Preliminary Design

A *Technical Drawing of the Preliminary Design* is a .slddwg file generated from the Preliminary Design Model. It must include:

- all standard orthographic views (front, top, side, bottom) and an isometric view,
- full dimensions, scale, title block, and any necessary annotations in line with drawing standards.

7.9 Technical Drawing of Detailed Design

A *Technical Drawing of the Detailed Design* is a .slddwg file created from the Detailed Design Model. It contains multiple pages and communicates both part and assembly features. It must include:

- **Page 1:** Exploded assembly view of parts with Balloons and a Bill of Materials (BOM), clearly arranged and numbered.
- **Pages 2–4:** Orthographic, sectional, and isometric views for **each part** used in the rover component (e.g. front, centre, and rear suspension links), showing all dimensions, title blocks, and annotations.

Note: Item 1 is not required for single-part components like the Wheel or Body.

7.10 Technical Drawing of Rover Sub-assembly

A *Technical Drawing of the Rover Sub-assembly* is a .slddwg file created from the Rover Sub-assembly Model. It includes:

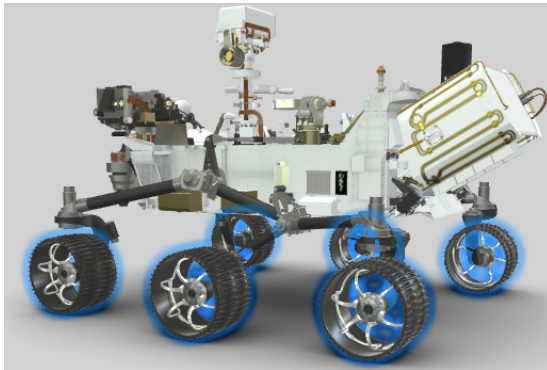
- **Page 1:** Exploded assembly view of rover components with Balloons and a BOM, arranged clearly.
- **Page 2:** Orthographic and isometric views of the rover sub-assembly with major dimensions only (e.g. height, width, length), scale, and title block.

7.11 Rover Assembly Drawing

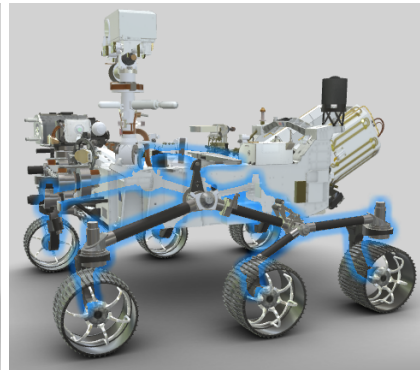
The *Rover Assembly Drawing* is a .slddwg file created from the final Rover Assembly Model. It includes multiple pages to communicate the complete rover design and must contain:

- **Page 1:** Exploded assembly view of the rover with Balloons and BOM, arranged properly.
- **Page 2:** Orthographic and isometric views of the entire rover with key dimensions (e.g. overall width, length, height with camera mast retracted), scale, and title block to confirm design compliance with requirements.

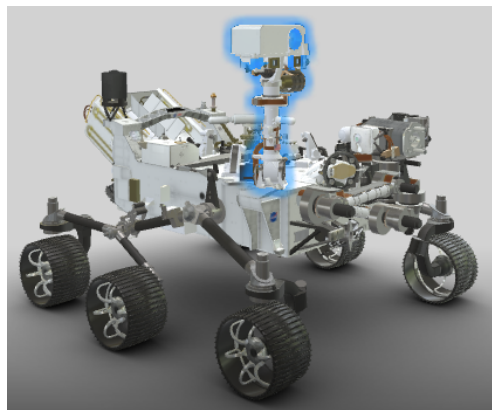
8 Appendix – 2: Rover Components



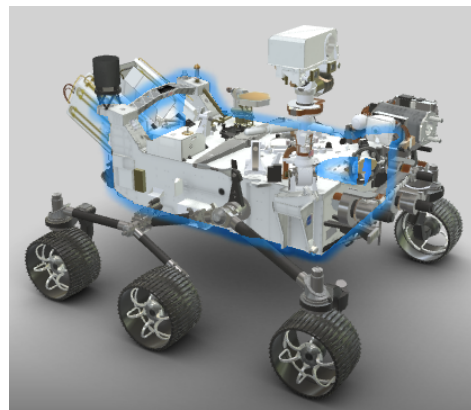
(1) Wheels



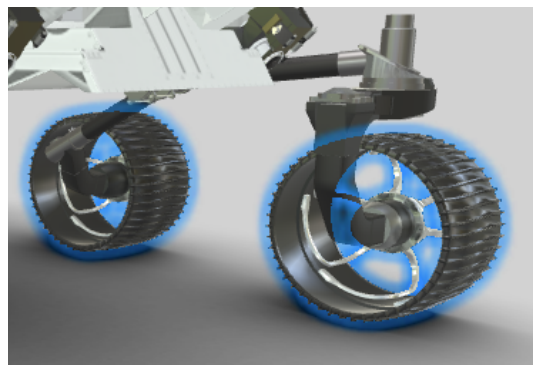
(2) Suspension



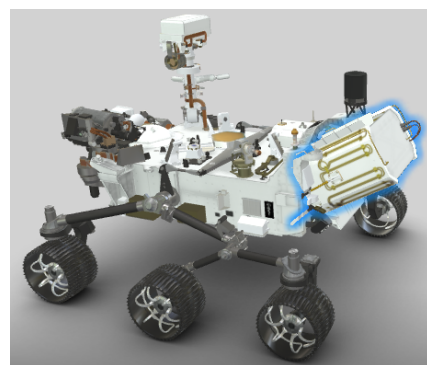
(3) Mast with camera



(4) Body



(5) Drive system



(6) Powerpack and Solar Panel

9 Appendix – 3 Group Work Contribution Table and Meeting Minutes Template

9.1 Part A Group Work Contribution Table

Fill this table and submit it in your individual design journal. Please provide this table as the first item in the individual design journal. Every team member must include this table in their design journal. It must be signed off by all members in the team.

Please note: Any student who attempts to modify or temper with the deadline or attendance data in this table will be given a 50% penalty for the Part A submission.

Table 1: Group work contribution table to be included in Part A submission

Group 84					
Name	Student No	Student Signature	Contribution		
			Part-A Preliminary Model Deadline	Part-A Detailed Model Deadline	No of Meetings Addedded (Wk 3 to 8)
John Smith	3013247		Met	Met	5
Jane Smith	3013259		Met	Met	4

9.2 Part B Group Work Contribution Table

Fill this table and submit it in your individual design journal. Please provide this table as the first item in the group design journal. It must be signed off by all members in the team.

Table 2a: Group work contribution table to be included in Part B submission

Group 84					
Name	Student No	Student Signature	Contribution		
			Part-A Preliminary Model Deadline	Part-A Detailed Model Deadline	No of Meetings Addedded (Wk 4 to 12)
John Smith	3013247		Met	Met	5
Jane Smith	3013259		Met	Met	4

Fill this table and update the items highlighted in yellow and submit it in your individual design journal. Please provide this table as the first item in the group design journal. It must be signed off by all members in the team.

Table 2b: Group work contribution table to be included in Part B submission

Name	Student ID	Contribution					
		Assembly Model	Assembly Exploded View	Assembly Animation	Assembly Drawing	VR Demo	Prepared PPT slides
John Smith	3013247	Yes	No	No	No	No	Yes
Jane Smith	3013259	Yes	No	No	No	No	Yes

9.3 Meeting Minutes Example

Example of the minutes for the meetings held between Week 3 to 8 for Part A of the project.

Week Number : 3			
Meeting Topic		Wheel Size	
Place		Face to face or Online	
Attendees			
Student ID	Name		Attendance
3013247	John	Smith	Yes
3013259	Jane	Smith	No

Agenda Details
We must finalise the wheel size. This will enable us to calculate the size of the power pack needed to complete the mission. It will also help finalise the overall height of the rover.

Decisions			
Action Plan	Responsibility	Deadline	Status
Watch the lectorial demonstrating the method for calculating the wheel size for completing the mission.	John, Jane	Fri, Wk 4	Complete
Calculate the wheel size according the instructions and compare cross check your finding.	John, Jane	Fri, Wk 4	Complete
Share the dimensions of the wheel with the team	John	Fri, Wk 4	Complete
Create a preliminary design of the electric motor and ensure it assembles with the preliminary design of the wheel	Jane	Wed, Wk 6	In progress
Create a preliminary design of the wheel	John	Fri, Wk 5	Complete
Make necessary changes to the design	John, Jane	Wed, Wk 6	In progress

9.4 Meeting Minutes Template

Fill this table with minutes for the meetings held between Week 3 to 8 for Part A of the project.

Week Number : 3			
Meeting Topic			
Place			
Attendees			
Student ID	Name		Attendance

Agenda Details

Decisions			
Action Plan	Responsibility	Deadline	Status

Decisions			
Action Plan	Responsibility	Deadline	Status